

C++ Programming Assignments

References, Pass by Reference, Pass by Address & Dynamic Memory Allocation

Assignment : Pass by Address and Reference

1. Write a function that accepts the addresses of two integers and swaps their values using pointers.
2. Write a function that accepts two integers using pass by reference and swaps their values.
3. Write a function using pass by reference to calculate the sum and difference of two numbers and return both results through reference parameters.
4. Write a function using pass by reference to reverse the digits of a number and update the original variable in main.

Assignment : Dynamic Memory Allocation – Arrays

1. Write a menu-driven C++ program to dynamically allocate an integer array of N elements and perform the following operations using user-defined functions:

1. Accept and display the array elements.
2. Find and display the largest and smallest elements.
3. Calculate and display the sum and average of the array elements.
4. Count and display the number of even and odd elements.
5. Search for a given element using linear search.
6. Exit the program.

2. Write a menu-driven C++ program to dynamically allocate memory for the prices of N products in a store and perform the following operations using user-defined functions:

1. Accept and display the **product prices**.
2. Find and display the **highest and lowest-priced products**.
3. Calculate and display the **total inventory value and average product price**.
4. Count the number of products whose price is **above and below the average price**.
5. Search for a particular **product price** using linear search.
6. Apply a given **discount percentage** to all product prices and display the updated prices.
7. Exit the program